


ITT
PRODUCT SPECIFICATION
IEC 62196 TYPE 2 COUPLER
 Electric Vehicle Connector and Equipment Plug

CAS21127E-CU
CUSTOMER


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1 Scope

This product specification describes connectors dedicated to an AC charging system acc. to IEC 62196, Type 2. The connectors are intended for use in an infrastructure described by IEC 61851. This may be with electric vehicles (EV), with charging stations or with service equipment (EVSE).

This specification doesn't describe the specific requirements for the infrastructure. Specification of Infrastructure could show requirements exceeding the requirements of IEC 62196-1 and IEC 62196-2 current revision.

The connectors are designed for Mode 2 and Mode 3, Level 2 AC charging systems for 1phase 250VAC and 3phase 480VAC. The connectors must not be used for Mode 4 DC charging systems. The connectors can be used for both indoor and outdoor applications.



ACI3-E2 Equipment plug
(EV plug)



ACI3-T2 vehicle connector



ACI3-J2 connection cable

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1.1 Intended Use Connectors have been designed and manufactured using state-of-the-art techniques and in accordance with approved safety regulations. Nevertheless, if not used as intended, risks for the user or third parties may still arise, or connectors and other material assets may be damaged. Intended use includes: - Application within the stated performance limits - Application when in good condition, without any damage - Application only in conjunction with designated vehicle inlets according to IEC 62196-1 & 2 - Application only with infrastructure acc. to IEC 62851-1				
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3 Part Number Configuration (Type Designation)

Type code designation		Example: ACI3-J2-032-1BK-025A-1A2	
AC	Product line	AC	AC Product line
I	Certification Code	I	EN 62196-1 / IEC 62196-1 EN 62196-2 / IEC 62196-2
3	Generation	3	Generation 3
-		-	Dash
T1	Type/ Jumper	E2	Tethered lead, mode 2 & 3, Case A, Type 2 Equipment Plug (EV Plug)
		T2	Tethered lead, mode 2 & 3, Case C, Type 2 Vehicle Connector
		J2	Connection cable, mode 3, Case B, Type 2 - Type 2
-		-	Dash
032	Amperage (3 digits)	020	20A
		032	32A
		063	63A (3 Phase only)
-		-	Dash
1	Phase/ Level	1	SINGLE PHASE (250VAC)
		3	THREE PHASE (480VAC)
BK	Color of housing	BK	Black (standard)
-		-	Dash
XXX	Cable length	XXX	LENGTH IN XXX x 0.1m (017 = 1.7m)
A	Cable type	A	Standard cable acc. to EN 50620
-		-	Dash
1	Contact System	1	Slotted female contacts and/or Pin Contacts
A	Sealing	A	IP rating acc. to Standard
1	Dust Cap	1	no dust cap supplied (E2 and T2 only)
		2	Rubber dust cap with PA rope
-		-	Dash
000	Mod Code*	000 to 999	Modification Code given by engineering. For Standard part number leave away these 3 digits and previous dash same mod code list will be used for all AC connectors starting with gen2.

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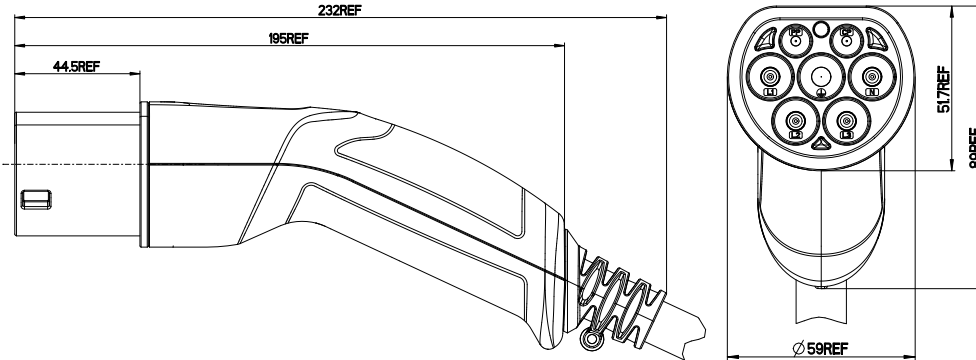

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IEC 62196 TYPE 2 COUPLER
 Electric Vehicle Connector and Equipment Plug

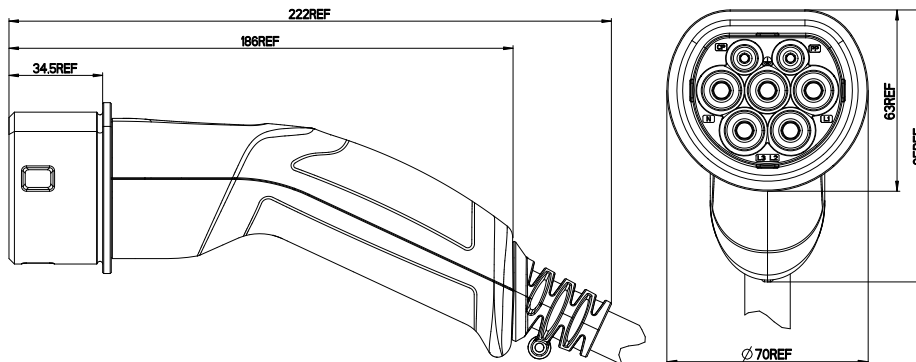
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3.1 Dimensions

3.1.1 Equipment Plug (EV Plug)



3.1.2 Vehicle Connector



3.2 Cable length

The signal transmission depends on the cable length and may be adversely affected. Therefore, the following is specified:

- The cable capacity must be assessed in the overall system of the charging station and must not exceed 3100 pF (IEC 61851-1:2017, Annex A, Table A.2, Note d).
- Interference-free V2G communication in accordance with ISO 15118 is not guaranteed for cable lengths over 10 m (ISO IEC 15118-3:2015, A.11.3, Table A.11).
- Cable management is required in certain regions if the cable length exceeds 5.0 m (Switzerland) or 7.5 m (USA) (IEC 61851-1:2017).

The maximum cable length is 15m, limited by manufacturing.

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3.3 Connector Weight

DESCRIPTION	CONNECTOR WEIGHT in g	CABLE WEIGHT PER m in g
ACI3-T2-020-1...	≈310	≈155
ACI3-T2-020-3...	≈340	≈235
ACI3-T2-032-1...	≈330	≈275
ACI3-T2-032-3...	≈380	≈430
ACI3-T2-063-3...	≈440	≈1030
ACI3-E2-020-1...	≈310	≈155
ACI3-E2-020-3...	≈340	≈235
ACI3-E2-032-1...	≈335	≈275
ACI3-E2-032-3...	≈380	≈430
ACI3-E2-063-3...	≈440	≈1030
ACI3-J2-020-1...	≈595	≈155
ACI3-J2-020-3...	≈655	≈235
ACI3-J2-032-1...	≈640	≈275
ACI3-J2-032-3...	≈730	≈430
ACI3-J2-063-3...	≈855	≈1030
Dust cap for ACI3-E2	≈20	
Dust cap for ACI3-T2	≈20	
Dust caps for ACI3-J2	≈40	

3.4 Visual Appearance

Default color of connector is black. Color of insulator (black), cable bending protection (black) and accessories as dust cap (black).

Standard handle parts are matt black with texture. Other colors are available on request only.



3.5 Internal Wiring

By default, Vehicle Connectors and EV Equipment Plugs will be provided with cable acc. to EN 50620:2020 (TPE insulated). Special cable for use on other countries are available on request.

Resistors ensure correct cable is used for different current rating and to determine different mating conditions.

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3.5.1 Cable Specification Matrix

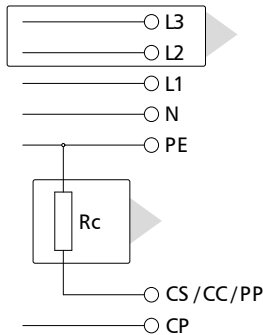
current rating	20A	32A	63A
POWER Wire Size	2.5 mm ²	6.0 mm ²	16.0 mm ²
PE Wire Size	2.5 mm ²	6.0 mm ²	16.0 mm ²
SIGNAL Wire Size	0.5 mm ²	0.5 mm ²	0.5 mm ²
CABLE OD 1phase	9.8 mm	12.2 mm	-
CABLE OD 3phase	12.4 mm	15.8 mm	22.5 mm
COLOR Cable Jacket	black	black	black
Cable Certification	EN 50620:2020	EN 50620:2020	EN 50620:2020

Quantity of signal wires is subject to change and depends on connector style. See customer use drawings and following wiring diagrams for reference.

3.5.2 Conductor Standard Colors

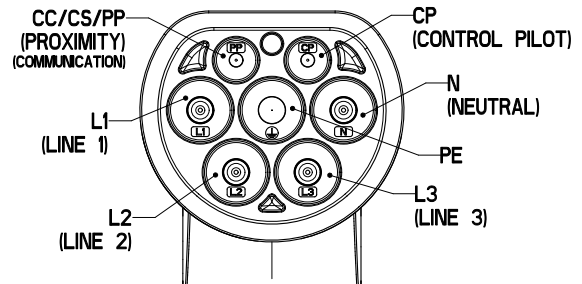
PIN Designation	Color
L1	BROWN
L2	BLACK
L3	GREY
N	BLUE
PE	GREEN/YELLOW
CS/CC/PP	RED
CP	WHITE

3.5.3 Equipment Plug

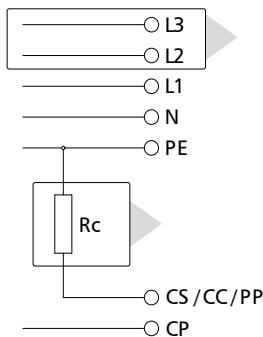


L2 and L3 for 3phase connectors only

Current Rating	Rc
20A	680Ω / 0.5W
32A	220Ω / 1.0W
63A	100Ω / 1.0W

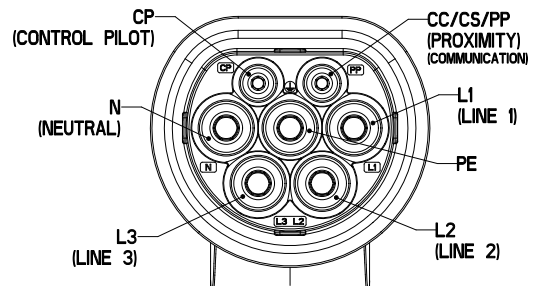


3.5.4 Vehicle Connector



L2 and L3 for 3phase connectors only

Current Rating	Rc
20A	680Ω / 0.5W
32A	220Ω / 1.0W
63A	100Ω / 1.0W



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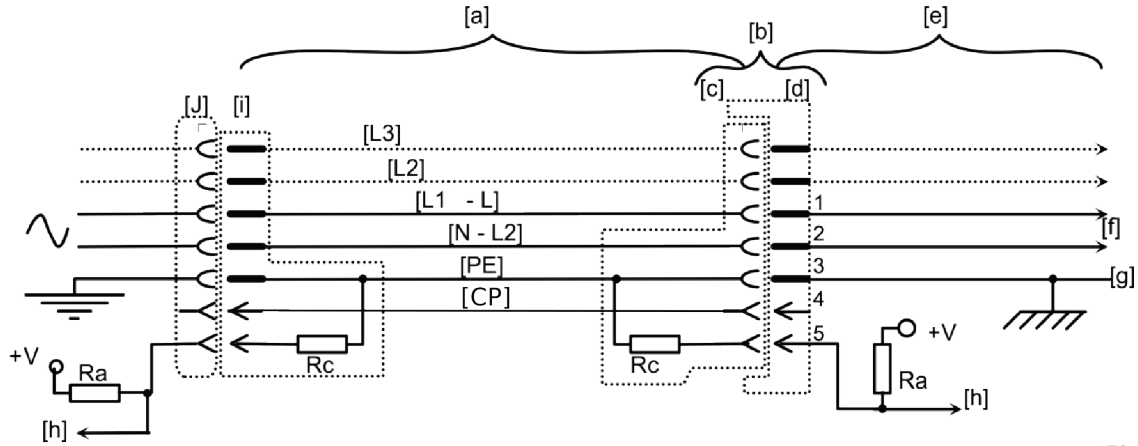
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3.6 Schematic Diagram of Typical Charging Connection acc. to IEC61851



IEC

KEY	DESCRIPTION
a	cable assembly
b	vehicle coupler
c	ACI3-T2 (Vehicle Connector)
d	vehicle inlet
e	circuit on the vehicle
f	AC supply to the vehicle
g	vehicle chassis connection
h	to proximity detection circuit
i	ACI3-E2 (EV plug, Equipment Plug)
J	EV socket-outlet
Rc	Current coding resistance

Based on IEC 61851-1:2017

4 Physical Characteristics

4.1 Materials

DESCRIPTION	MATERIAL	FINISH
contact pin	brass	Silver plated
contact socket	brass	Silver plated
insulator front, contact holding plate	PA, glass filled	
O-ring, sealing elements	EPDM rubber	
strain relief	aluminum	
handle, housing	polybutylene terephthalate blend (PBT/PC)	
Bending protection	thermoplastic elastomer TPV	
bolts	stainless steel	
cable	copper / rubber or TPE	
PCB assembly	PCB board, metal oxide resistor, copper-conductor, XLPE	
Cable tie	PA	
Dust cap	EPDM rubber, PA, Cu, Polyolefin	

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4.2 Storage Temperatures and Humidity Conditions During shipping and storing, -40°C to +80°C is acceptable. If condensation is expected to be above average, additional measures should be taken. Acceptable humidity depends on temperature. When the highest temperature is +40°C, the air relative humidity shall not exceed 50%. When the highest temperature is +20°C, the relative humidity can reach 90% for a short period.														
4.3 Operating Temperatures Connectors are intended for normal use at rated current depending on certified standard like shown in table below:														
<table><tr><th>Standard</th><th>Temperature limits</th></tr><tr><td>DIN EN 62196-1:2015</td><td>-30°C to +50°C</td></tr><tr><td>DIN EN 62196-2:2017</td><td>-30°C to +50°C</td></tr><tr><td>IEC 62196-1:2022</td><td>-30°C to +40°C</td></tr><tr><td>IEC 62196-2:2022</td><td>-30°C to +40°C</td></tr></table>					Standard	Temperature limits	DIN EN 62196-1:2015	-30°C to +50°C	DIN EN 62196-2:2017	-30°C to +50°C	IEC 62196-1:2022	-30°C to +40°C	IEC 62196-2:2022	-30°C to +40°C
Standard	Temperature limits													
DIN EN 62196-1:2015	-30°C to +50°C													
DIN EN 62196-2:2017	-30°C to +50°C													
IEC 62196-1:2022	-30°C to +40°C													
IEC 62196-2:2022	-30°C to +40°C													
In range from +40 °C to +50°C a reduction of operating current might be required to prevent overheating. Necessity needs to be determined by customer in final application.														
4.4 IP Rating EV plug and vehicle connector with dust cap mounted reach IP44. EV plug and vehicle connector in mated condition with inlets or outlets acc. to IEC62196 reach IP44. EV plug and vehicle connector without dust cap are marked with IP-Rating IP24 on connector as required by IEC 61851-1. For EV plug and vehicle connector without dust cap an appropriate parking position to ensure minimum protection IP24 according to IEC 61851-1 is required. Connection cable is supplied with dust cap on both sides. Connection Cable is marked IP44.														
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5 Electrical Characteristics

Electrical tests will be performed acc. to DIN EN 62196-1:2015-06 (VDE 0623-5-1:2015-06)

5.1 Operating Voltage and Rated Current

1phase connectors are intended for operating voltage up to 250VAC. Connectors rated 20A, 32A are available.

3phase connectors are intended for operating voltage up to 480VAC. Connectors rated 20A, 32A and 63A are available.

5.2 Clearance and Creepage

Connectors are meant for overvoltage category II, pollution degree 3 up to operating height of 4000m.

Acc. to operating place other values could be necessary to consider.

Path	Clearance in mm	Creepage in mm
Contact-Contact	13.5	16
Contact-Touchable Surfaces	13	17

5.3 Insulation Resistance

Test acc. to DIN EN 62196-1:2015-06 (VDE 0623-5-1:2015-06) §21.2

With a mated pair of connector assemblies, measurement of the insulation resistance shall be more than 5 MΩ.

Test acc. to IEC 62196-1:2022 §21.2

With a mated pair of connector assemblies, measurement of the insulation resistance shall be more than 5 MΩ.

5.4 Dielectric Strength

Test acc. to DIN EN 62196-1:2015-06 (VDE 0623-5-1:2015-06) §21.3

With a mated pair of connector assemblies, no flashover or breakdown shall occur during the test.

Test acc. to IEC 62196-1:2022 §21.3

With a mated pair of connector assemblies, no flashover or breakdown shall occur during the test.

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5.5 Temperature Rise

Test acc. to DIN EN 62196-1:2015-06 (VDE 0623-5-1:2015-06) §24 and DIN EN 62196-1:2015-06 (VDE 0623-5-1:2015-06) §16.5

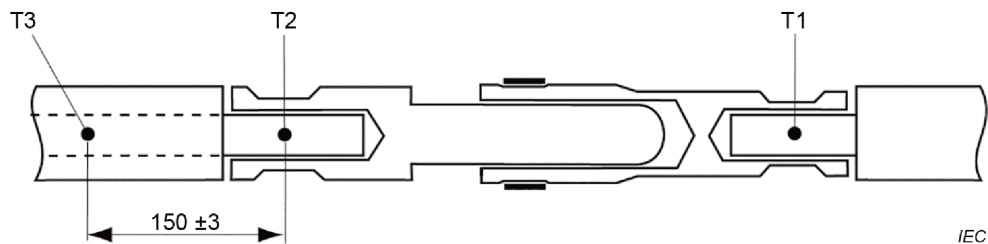
Sensors are located acc. to IEC62196 and additional sensors at location 1, 2, 3 at hottest location in these areas.

Test acc. to IEC 62196-1:2022 §24 and IEC 62196-1:2022 §16.5

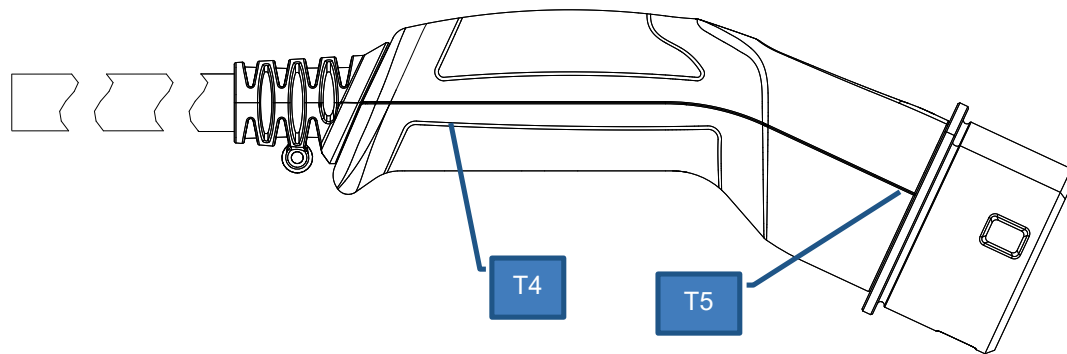
Sensors are located acc. to IEC62196 and additional sensors at location 1, 2, 3 at hottest location in these areas.

Measurement Point	Maximum temperature in °C Maximum temperature rise in K
Contact crimp T1 and T2	50K
Cable jacket temperature T3	90°C
Touchable metal parts	60°C
Graspable metal parts	50°C
Touchable non-metal parts	85°C
Graspable non-metal parts	60°C

Rated Current in A	Test Current in A	Wire Size in mm²
20	22	2.5
32	42	6.0
63	63	16.0



Based on IEC 62196-1:2022



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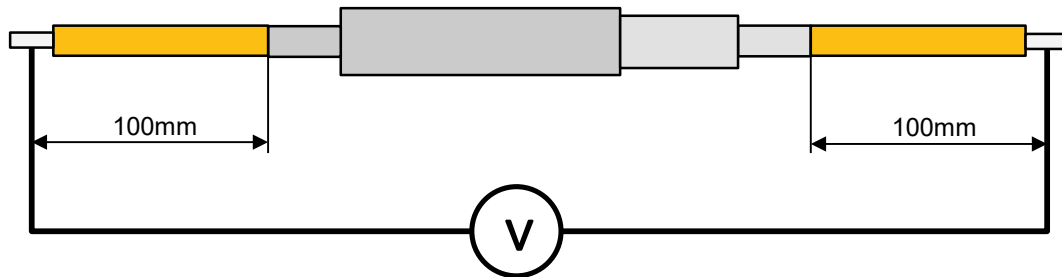
PRODUCT SPECIFICATION
IEC 62196 TYPE 2 COUPLER
 Electric Vehicle Connector and Equipment Plug

CAS21127E-CU
CUSTOMER
5.6 Low Level Contact Resistance

Measurements shall be performed using a four-wire method with a test current at rated current. Maximum contact resistance after environmental exposure shall not exceed 20 mΩ.

Maximum contact resistance initial state shall not exceed values given in tabulation below.

- a) Test Voltage – ≤20 mVDC
- b) Test Current – rated current



CONTACT STYLE	CURRENT RATING	VOLTAGE DROP in mV/A (mΩ)
6 mm (power, protective earth)	20A	≤ 1
	32A	≤ 2
	63A	≤ 3
3 mm (signal)		≤ 5

Given values are in initial state.

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6 Mechanical Characteristics				
6.1 Mating / Un-mating Forces				
Test acc. to DIN EN 62196-1:2015-06 (VDE 0623-5-1:2015-06) §16.16				
The force to mate and un-mate a coupler pair shall not exceed 100 N.				
Test acc. to IEC 62196-1:2022 §16.16				
The force to mate and un-mate a coupler pair shall not exceed 100 N.				
6.2 Contact Retention Force				
An axial force of 100 N per contact should not damage holding plate and should not cause contact to be moved more than 1.2 mm in axial direction. Force is applied on 1 contact after another and not at the same time.				
Dimensional reference is taken at a preload of 1 N				
6.3 Locking Mechanisms and Safety Devices				
IEC type 2 connectors use different locking and safety features to prevent inadvertent decoupling. Coupler will be locked in place by inlet / outlet.				
Control circuit using the CP contacts ensures all connectors are fully engaged during charging cycle.				
Vehicle coupler and Equipment Plugs will not be provided with mechanical locking latch.				
Test acc. to DIN EN 62196-1:2015-06 (VDE 0623-5-1:2015-06) §14				
Loaded with a 300 N force the connection shall not break, and the locking device shall not be damaged.				
6.4 Durability, Normal Operation				
Test acc. to DIN EN 62196-1:2015-06 (VDE 0623-5-1:2015-06) §23.1 and §23.3				
The coupler shall be designed to a minimum of 10k cycles of mechanical operation. Upon manual mating and un-mating of the connector assemblies, the mating forces must not increase to the point that normal human has difficulty mating them. After cycling, the devices shall not show any wear or deterioration that would impair further use.				
Test acc. to IEC 62196-1:2022 §23.1 and §23.3				
The coupler shall be designed to a minimum of 10k cycles of mechanical operation with dipping connector in a solution of sand and salt each 1k cycles. Upon manual mating and un-mating of the connector assemblies, the mating forces must not increase to the point that normal human has difficulty mating them. After cycling, the devices shall not show any wear or deterioration that would impair further use.				
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7 Environmental Characteristics				
7.1 Humidity Test acc. to IEC 62196-1:2022 §35 After an initial temperature rise test, connectors are exposed to humidity cycles and mating cycles without sand and salt. Humidity Cycles are followed by a final temperature rise test. Connector should not show excessive increase of overheating.				
7.2 High Temperature Life Test Test acc. to DIN EN 62196-1:2015-06 (VDE 0623-5-1:2015-06) §15 and §29 Rubber and plastic materials shall not show cracks, sealing components should not flow away and should not undergo any change impairing further use. Test acc. to IEC 62196-1:2022 §15 and §29 Rubber and plastic materials shall not show cracks, sealing components should not flow away and should not undergo any change impairing further use.				
7.3 Salt Spray Mist Test acc. to DIN EN 60068-2-11:2020 With the connectors mated, the test should be conducted under the following conditions: a) Salt concentration 5% b) Test duration 48 Hours c) Test conditions: Atomization d) Temperature: 35°C Test acc. to IEC 62196-1:2022 §30 Ferrous parts shall not show rust after cleaning with ethyl acetone, acetone, methyl ethyl ketone or an equivalent degreasing agent and immersion to 10% solution of ammonium chloride in water.				
7.4 Thermal Cycling Test acc. to IEC 62196-1:2022 §34 After an initial temperature rise test, connectors are exposed to thermal cycling test. Thermal cycling is followed by a final temperature rise test. Connector should not show excessive increase of overheating.				
8 Accessories				
8.1 Dust Cap For available dust caps see talking dog and customer use drawings.				
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	PRODUCT SPECIFICATION IEC 62196 TYPE 2 COUPLER Electric Vehicle Connector and Equipment Plug	CAS21127E-CU CUSTOMER
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9 Product Compliance Certificate

Certificate is depending on type designation:

Standard	Type	Certificate	ACI3-E2	ACI3-T2	ACI3-J2
DIN EN 62196-1:2015	VDE Mark	40056647 (ACI3-T2)	020-1	020-1	020-1
DIN EN 62196-2:2017		40056700 (ACI3-E2)	020-3	020-3	020-3
		40056775 (ACI3-J2)	032-1	032-1	032-1
			032-3	032-3	032-3
			063-3		
IEC 62196-1:2022	CB scheme	DE1-67408 (ACI3-T2)	032-1	032-1	032-1
IEC 62196-2:2022		DE1-67572 (ACI3-E2)	032-3	032-3	032-3
		DE1-67627 (ACI3-J2)		063-3	

<https://www2.vde.com/en/Institute/OnlineService/VDE-approved-products/Pages/Online-Search.aspx>

Search without reference number:

Search field	Value
Company	ITT Cannon GmbH
Type	ACI3

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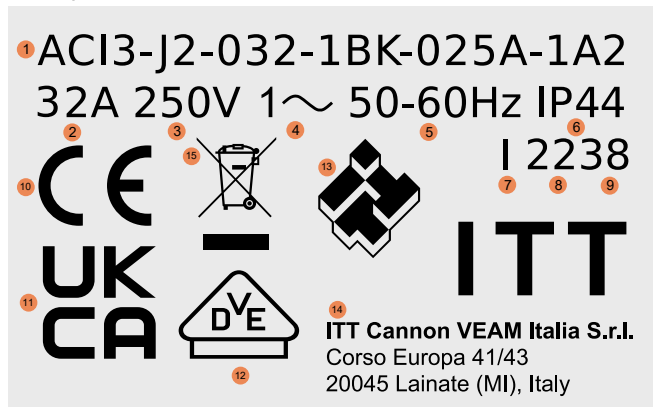
	PRODUCT SPECIFICATION IEC 62196 TYPE 2 COUPLER Electric Vehicle Connector and Equipment Plug	CAS21127E-CU CUSTOMER
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10 Declaration of Conformity

10.1 Certification Markings and Identification

Connectors and plugs are marked with VDE certification marks and connector rating as well as tracing information.

Example:



Marking	Description
1	type designation
2	Current Rating (max. operating current)
3	Voltage Rating (max. operating voltage)
4	Operating frequency
5	Count of phases
6	Protection class
7	Production site identification
8	Production year
9	Production week
10	CE-Mark
11	UKCA-Mark (UK Conformity Assessed)
12	VDE Certification mark (if available)
13	ITT Logo
14	Company address
15	WEEE logo (J2 version only)



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	PRODUCT SPECIFICATION IEC 62196 TYPE 2 COUPLER Electric Vehicle Connector and Equipment Plug	CAS21127E-CU CUSTOMER
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10.2 RoHS

Connector fulfils / fulfil the requirements of Directive 2011/65/EU (RoHS) and all amending Directives.
For official statement consult factory.

10.3 CE Declaration of Conformity

Standard	Document number
DIN EN 62196-1:2015	05-2023_EC
DIN EN 62196-2:2017	
IEC 62196-1:2022	07-2023_EC
IEC 62196-2:2022	

10.4 UKCA Declaration of Conformity

Standard	Document number
DIN EN 62196-1:2015	06-2023_UK
DIN EN 62196-2:2017	
IEC 62196-1:2022	08-2023_UK
IEC 62196-2:2022	

11 Document Information

11.1 Reference Source, Works Cited

- IEC. (2017, 07). IEC61851. *Electric vehicle conductive charging system - Part 1: General requirements*. Switzerland.
- IEC. (2022, 05). IEC62196-1. *Plugs, socket-outlets, vehicle connectors and vehicle inlets – Conductive charging of electric vehicles – Part 1: General requirements*. Switzerland.

11.2 Document Revision Record

Revision	Date	Comments
-	05.06.2023	first release

*****END*****

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